

1. Three balls are to be randomly selected without replacement from an urn containing 20 balls numbered 1 through 20. If we bet that at least one of the balls that are drawn has a number as large as or larger than 17, what is the probability that we win the bet?

Ans[0.508]

2. Independent trials consisting of the flipping of a coin having probability p of coming up heads are continually performed until either a head occurs or a total of n flips is made. If we let X denote the number of times the coin is flipped, then find out the probability distribution of X .
3. The probability mass function of a random variable X is given by $p(i) = \frac{e\lambda^i}{i!}$ for $i = 0, 1, 2, \dots$, where λ is some positive value. Find (a) $P(X = 0)$ and (b) $P(X > 2)$.
4. A contestant on a quiz show is presented with two questions, questions 1 and 2, which he is to attempt to answer in some order he chooses. If he decides to try question i first, then he will be allowed to go on to question j , $j \neq i$, only if his answer to question i is correct. If his initial answer is incorrect, he is not allowed to answer the other question. The contestant is to receive V_i dollars if he answers question i correctly, $i = 1, 2$. For instance, he will receive $V_1 + V_2$ dollars if he answers both questions correctly. If the probability that he knows the answer to question i is P_i , $i = 1, 2$, which question

should he attempt to answer first so as to maximize his expected winnings? Assume that the events E_i , $i = 1, 2$, that he knows the answer to question i are independent events.

5. The distribution function of a random variable X is

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } x > b. \end{cases}$$

Find $E(X)$ and $\text{Var}(X)$.

6. If distribution function of a random variable is $F(x) = 1 - e^{-2x}$, for $x > 0$, find $P(2 \leq X \leq 5)$.

7. Suppose that 15 percent of the families in a certain community have no children, 20 percent have 1 child, 35 percent have 2 children, and 30 percent have 3. Suppose further that in each family each child is equally likely (independently) to be a boy or a girl. If a family is chosen at random from this community, then B , the number of boys, and G , the number of girls, in this family. Find out the joint probability mass function of B and G .

8. If joint pdf of (X, Y) is $f(x, y) = x^2 + \frac{xy}{3}$ for $0 < x < 1$, $0 < y < 2$. Find $P(Y \leq \frac{1}{2})$.
9. Suppose that X is a random variable that takes on values 0, 2 and 3 with probabilities 0.3, 0.1, 0.6 respectively. Let $Y = 3(X - 1)^2$.
- (a) What is the expectation of X ? (b) What is the expectation of Y ?

10. Suppose that the cdf of X is given by: $F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{1}{4} & \text{for } 0 \leq x < 2 \\ \frac{1}{2} & \text{for } 2 \leq x < 4 \\ 1 & \text{for } x \geq 4. \end{cases}$

Determine the pmf of X.

11. Let X be a random variable with cdf $F(x) = 2x^2 - x^4$, for $0 \leq x \leq 1$.

(a) Compute $P(\frac{1}{4}X \leq \frac{3}{4})$.

(b) What is the pdf of X?

12.

	Y=1	Y=2	Y=3	
X=1	$\frac{1}{6}$	0	...	$\frac{1}{3}$
X= 2	...	$\frac{1}{4}$	...	$\frac{1}{3}$
X=3	...	...	$\frac{1}{4}$	...
	$\frac{1}{6}$	$\frac{1}{3}$	...	1

Complete the table.

13. An investment firm offers its customers municipal bonds that mature after varying numbers of years. Given that the cumulative distribution function of T, the number of years to maturity for a randomly selected bond, is

$$F(t) = \begin{cases} 0 & \text{if } t < 1; \\ \frac{1}{4} & \text{if } 1 \leq x < 3; \\ \frac{1}{2} & \text{if } 3 \leq x < 5; \\ \frac{3}{4} & \text{if } 5 \leq x < 7; \\ 1 & \text{if } x \geq 7. \end{cases}$$

Find $P(T \leq 5 | T \geq 2)$

14. The rv X has the mgf $M(t) = \frac{1}{6}e^{-2t} + \frac{1}{3}e^{-t} + \frac{1}{4}e^t + \frac{1}{4}e^{2t}$. Compute $P(|X| \leq 1)$.

15. A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let X and Y , respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & \text{if } 0 \leq x \leq 1; 0 \leq y \leq 1 ; \\ 0 & \text{otherwise.} \end{cases}$$

Find $P[(X, Y) \in A]$, where $A = \{(x, y) | 0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2}\}$

16. The joint density for the random variables (X, Y) , where X is the unit temperature change and Y is the proportion of spectrum shift that a certain atomic particle produces, is

$$f(x, y) = \begin{cases} 10xy^2 & \text{if } 0 < x < y < 1; \\ 0 & \text{otherwise.} \end{cases}$$

Find $P(Y > \frac{1}{2} | x = 0.25)$